

Capturing from Adapter for loopback traffic capture

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 9500

No.	Time	Source	Destination	Protocol	Length	Info
20	41.158671100	127.0.0.1	127.0.0.1	TCP	44	53870 → 9500 [ACK] Seq=1 Ack=1 Win=2619648 Len=0
38	227.709473500	127.0.0.1	127.0.0.1	TCP	63	53870 → 9500 [PSH, ACK] Seq=1 Ack=1 Win=2619648 Len=19
39	227.709530800	127.0.0.1	127.0.0.1	TCP	44	9500 → 53870 [ACK] Seq=1 Ack=20 Win=2619648 Len=0
53	417.022335300	127.0.0.1	127.0.0.1	TCP	47	53870 → 9500 [PSH, ACK] Seq=20 Ack=1 Win=2619648 Len=3
54	417.022420000	127.0.0.1	127.0.0.1	TCP	44	9500 → 53870 [ACK] Seq=1 Ack=23 Win=2619648 Len=0
55	417.022594800	127.0.0.1	127.0.0.1	TCP	44	53870 → 9500 [FIN, ACK] Seq=23 Ack=1 Win=2619648 Len=0
56	417.022626000	127.0.0.1	127.0.0.1	TCP	44	9500 → 53870 [ACK] Seq=1 Ack=24 Win=2619648 Len=0

Frame 38: Packet, 63 bytes on wire (504 bits), 63 bytes captured (504 bits) on interface \Device\NPF{...} Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 53870, Dst Port: 9500, Seq: 1, Ack: 1, Len: 19

Data (19 bytes)

0000 02 00 00 00 45 00 00 3b 1b 09 40 00 80 06 00 00E.;.@.....
0010 7f 00 00 01 7f 00 00 01 d2 6e 25 1c fe 80 55 ffn%..U..
0020 c4 78 4f 97 50 18 27 f9 5c 99 00 00 54 68 72 65 ..xO.P.'.\...Thre
0030 65 20 57 61 79 20 48 61 6e 64 53 68 61 6b 65 ..e Way Ha ndShake

wireshark_Adapter for loopback traffic capture18NXL3.pcapng Packets: 56 · Displayed: 14 (25.0%) Profile: Default

Word 2016 powershell - TP - Visual Studio Code

```
onsole Client
-U "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client
Console Client
fin
Fin du programme
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -U "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client.py"
Console Client
Three Way HandShake
FIN
Fin du programme
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP>
```

Wireshark

FichierSorti...

ENG FR 3:39 PM 2/27/2026

Capturing from Adapter for loopback traffic capture

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 9500

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	47	64588 → 9500 [PSH, ACK] Seq=1 Ack=1 Win=10233 Len=3
2	0.000060300	127.0.0.1	127.0.0.1	TCP	44	9500 → 64588 [ACK] Seq=1 Ack=4 Win=10233 Len=0
3	0.000163400	127.0.0.1	127.0.0.1	TCP	44	64588 → 9500 [FIN, ACK] Seq=4 Ack=1 Win=10233 Len=0
4	0.000195100	127.0.0.1	127.0.0.1	TCP	44	9500 → 64588 [ACK] Seq=1 Ack=5 Win=10233 Len=0
13	21.497608300	127.0.0.1	127.0.0.1	TCP	44	9500 → 64588 [RST, ACK] Seq=1 Ack=5 Win=0 Len=0
18	41.158537000	127.0.0.1	127.0.0.1	TCP	56	53870 → 9500 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
19	41.158622700	127.0.0.1	127.0.0.1	TCP	56	9500 → 53870 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
20	41.158671100	127.0.0.1	127.0.0.1	TCP	44	53870 → 9500 [ACK] Seq=1 Ack=1 Win=2619648 Len=0

Frame 38: Packet, 63 bytes on wire (504 bits), 63 bytes captured (504 bits) on interface \Device\NPF{...} Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 53870, Dst Port: 9500, Seq: 1, Ack: 1, Len: 19

Data (19 bytes)

0000 02 00 00 00 45 00 00 3b 1b 09 40 00 80 06 00 00E.;@.....
0010 7f 00 00 01 7f 00 00 01 d2 6e 25 1c fe 80 55 ffn%.U.....
0020 c4 78 4f 97 50 18 27 f9 5c 99 00 00 54 68 72 65 ..xU.P....\....thre
0030 65 20 57 61 79 20 48 61 6e 64 53 68 61 6b 65e Way Ha ndShake

wireshark_Adapter for loopback traffic capture18NXL3.pcapng

Packets: 49 · Displayed: 10 (20.4%)

Profile: Default

python - TP - Visual Studio Code

```
onsole Client
-U "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client
Console Client
fin
Fin du programme
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -U "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client.py"
Console Client
Three Way HandShake
```

Word 2016

Telegram

Recycle Bin

Wireshark

FichierSorti...

ENG FR 3:37 PM 2/27/2026

Capturing from Adapter for loopback traffic capture

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 9500

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	47	64588 → 9500 [PSH, ACK] Seq=1 Ack=1 Win=10233 Len=3
2	0.000060300	127.0.0.1	127.0.0.1	TCP	44	9500 → 64588 [ACK] Seq=1 Ack=4 Win=10233 Len=0
3	0.000163400	127.0.0.1	127.0.0.1	TCP	44	64588 → 9500 [FIN, ACK] Seq=4 Ack=1 Win=10233 Len=0
4	0.000195100	127.0.0.1	127.0.0.1	TCP	44	9500 → 64588 [ACK] Seq=1 Ack=5 Win=10233 Len=0
13	21.497608300	127.0.0.1	127.0.0.1	TCP	44	9500 → 64588 [RST, ACK] Seq=1 Ack=5 Win=0 Len=0
18	41.158537000	127.0.0.1	127.0.0.1	TCP	56	53870 → 9500 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
19	41.158622700	127.0.0.1	127.0.0.1	TCP	56	9500 → 53870 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
20	41.158671100	127.0.0.1	127.0.0.1	TCP	44	53870 → 9500 [ACK] Seq=1 Ack=1 Win=2619648 Len=0

Frame 1: Packet, 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface \Device Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 64588, Dst Port: 9500, Seq: 1, Ack: 1, Len: 3

Data (3 bytes)

0000 02 00 00 00 45 00 00 2b 1a e7 40 00 80 06 00 00E + + @.....
0010 7f 00 00 01 7f 00 00 01 fc 4c 25 1c 09 45 9b d0L%..E..
0020 b7 d6 9a 5c 50 18 27 f9 9c b2 00 00 66 69 6e\P..'.----fin

Packets: 34 · Displayed: 8 (23.5%) Profile: Default

ENG 3:34 PM
FR 2/27/2026

Capturing from Adapter for loopback traffic capture

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 9500 || udp.port == 9500

No.	Time	Source	Destination	Protocol	Length	Info
122	4.045469600	127.0.0.1	127.0.0.1	TCP	44	9500 → 53163 [RST] Seq=1234567890 Win=0 Len=0
127	4.549049800	127.0.0.1	127.0.0.1	TCP	56	[TCP Port number 9500] → 53163 [RST] Seq=1234567890 Win=0 Len=0
128	4.549075800	127.0.0.1	127.0.0.1	TCP	44	9500 → 53163 [RST] Seq=1234567890 Win=0 Len=0
129	5.049695900	127.0.0.1	127.0.0.1	TCP	56	[TCP Port number 9500] → 53163 [RST] Seq=1234567890 Win=0 Len=0
130	5.049732800	127.0.0.1	127.0.0.1	TCP	44	9500 → 53163 [RST] Seq=1234567890 Win=0 Len=0
131	5.564700200	127.0.0.1	127.0.0.1	TCP	56	[TCP Port number 9500] → 53163 [RST] Seq=1234567890 Win=0 Len=0
132	5.564722800	127.0.0.1	127.0.0.1	TCP	44	9500 → 53163 [RST] Seq=1234567890 Win=0 Len=0
133	21.546455900	127.0.0.1	127.0.0.1	TCP	56	53164 → 9500 [RST] Seq=1234567890 Win=0 Len=0
134	21.546479000	127.0.0.1	127.0.0.1	TCP	44	9500 → 53164 [RST] Seq=1234567890 Win=0 Len=0
135	22.047188400	127.0.0.1	127.0.0.1	TCP	56	[TCP Port number 9500] → 53164 [RST] Seq=1234567890 Win=0 Len=0
136	22.047233100	127.0.0.1	127.0.0.1	TCP	44	9500 → 53164 [RST] Seq=1234567890 Win=0 Len=0
141	22.549505200	127.0.0.1	127.0.0.1	TCP	56	[TCP Port number 9500] → 53164 [RST] Seq=1234567890 Win=0 Len=0
142	22.549535700	127.0.0.1	127.0.0.1	TCP	44	9500 → 53164 [RST] Seq=1234567890 Win=0 Len=0
143	23.065421300	127.0.0.1	127.0.0.1	TCP	56	[TCP Port number 9500] → 53164 [RST] Seq=1234567890 Win=0 Len=0
144	23.065458800	127.0.0.1	127.0.0.1	TCP	44	9500 → 53164 [RST] Seq=1234567890 Win=0 Len=0
145	23.565988500	127.0.0.1	127.0.0.1	TCP	56	[TCP Port number 9500] → 53164 [RST] Seq=1234567890 Win=0 Len=0
146	23.566033200	127.0.0.1	127.0.0.1	TCP	44	9500 → 53164 [RST] Seq=1234567890 Win=0 Len=0

Frame 119: Packet, 56 bytes on wire (448 bits) 0000 02 00 00 00 45 00 00 34 1a b6 40 00 80 00 00 00 00 00 01 7f 00 00 01 cf ab 25 1c 53 80 00 00 00 00 80 02 ff ff a3 09 00 00 02 00 00 00 01 03 03 08 01 01 04 02

Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Destination: 127.0.0.1

Transmission Control Protocol, Src Port: 53164, Dst Port: 9500

wireshark_Adapter for loopb...raffic capture\TRL3.pcapn Packets: 146 · Displayed: 20 (13.7%) Profile: Default

powershell - TP - Visual Studio Code

Oops, something went wrong. Please report this bug with the details below.
Report on GitHub: <https://github.com/lzybkr/PSReadLine/issues/new>

Last 1 Keys:
UpArrow ...

```

at Microsoft.PowerShell.PSConsoleReadLine.ForceRender()
at Microsoft.PowerShell.PSConsoleReadLine.HistoryRecall(Int32 direction)
at Microsoft.PowerShell.PSConsoleReadLine.PreviousHistory(Nullable`1 key, Object arg)
at Microsoft.PowerShell.PSConsoleReadLine.ProcessOneKey(ConsoleKeyInfo key, Dictionary`2 dispatchTable, Boolean ignoreIfNoAction, Object arg)
at Microsoft.PowerShell.PSConsoleReadLine.InputLoop()
at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

```

PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -U "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client.py"

Console Client

Traceback (most recent call last):

File "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client.py", line 7, in <module>

ConnexionAUnServeur.connect((host, port)) # connexion avec le serveur (envoi e flag SYN)

ConnectionRefusedError: [WinError 10061] No connection could be made because the target machine actively refused it

PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -U "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client.py"

Console Client

Traceback (most recent call last):

File "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Client.py", line 7, in <module>

ConnexionAUnServeur.connect((host, port)) # connexion avec le serveur (envoi e flag SYN)

ConnectionRefusedError: [WinError 10061] No connection could be made because the target machine actively refused it

PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP>

```
python - TP - Visual Studio Code
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -u
"c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Serveur.py"

Lancement serveur
Connexion de la machine = ('127.0.0.1', 58487)
Je suis un Client écrit en Java
```

```
javac - TP - Visual Studio Code
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP>
Oops, something went wrong. Please report this bug with the details below.
Report on GitHub: https://github.com/PowerShell/PSReadline/issues/new
```

```
Last 1 Keys:
UpArrow
```

```
Exception:
System.ArgumentOutOfRangeException: The value must be greater than or equal to zero
and less than the console's buffer size in that dimension.
Parameter name: top
Actual value was -2.
   at System.Console.SetCursorPosition(Int32 left, Int32 top)
   at Microsoft.PowerShell.PSConsoleReadLine.ReallyRender(RenderData renderData,
String defaultColor)
   at Microsoft.PowerShell.PSConsoleReadLine.ForceRender()
   at Microsoft.PowerShell.PSConsoleReadLine.HistoryRecall(Int32 direction)
   at Microsoft.PowerShell.PSConsoleReadLine.PreviousHistory(Nullable`1 key, Object
arg)
   at Microsoft.PowerShell.PSConsoleReadLine.ProcessOneKey(ConsoleKeyInfo key, Dictionary`2 dispatchTable, Boolean ignoreIfNoAction, Object arg)
   at Microsoft.PowerShell.PSConsoleReadLine.InputLoop()
   at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)
-----
```

```
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> cd "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\" ; if ($?) { java
c Client.java } ; if ($?) { java Client }
Lancement de l'application Client DESKTOP-VNSJEKR/127.0.0.1
```

Input

?

Entrez le message à transmettre:

OK Cancel

```
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -u  
"c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Serveur.py"
```

Lancement serveur

Connexion de la machine = ('127.0.0.1', 58487)

□

```
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP>  
Oops, something went wrong. Please report this bug with the details below.  
Report on GitHub: https://github.com/lzybkr/PSReadLine/issues/new
```

Last 1 Keys:

UpArrow

Exception:

System.ArgumentOutOfRangeException: The value must be greater than or equal to zero and less than the console's buffer size in that dimension.

Parameter name: top

Actual value was -2

at System.Console.WriteLine(Int32 top)
at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

String defaultPrompt = "PS ";
at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

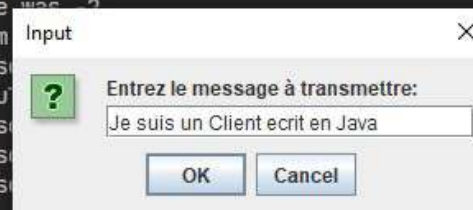
at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)

PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> cd "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\" ; if (\$?) { java

c Client.java } ; if (\$?) { java Client }
Lancement de l'application Client DESKTOP-VNSJEKR/127.0.0.1
□



File Edit Selection View Go Run ... TP

Server.py Client.py test Client.jav Extension Extension

test.py

```
1 """
2 |---Réponse---
3 Quand 2 Clients envoient des messages au meme serveur le serveur
4 se deconnecte du premier puis se connecte au deuxieme et recoit le
5 dernier message envoyé seulement.
6 """
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code python Code

```
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> re06\Reseaux\TP\Se
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Rea\TP> python -u "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -u "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -u "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP> python -u "c:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Reseaux\TP\Serveur.py"
Lancement serveur
Connexion de la machine = ('127.0.0.1', 64210)
fin
Deconnexion de : ('127.0.0.1', 64210)
Connexion de la machine = ('127.0.0.1', 60338)
Message2
[]
```

main* 01:61 0 3 Indexing completed. Java: Ready Ln 2, Col 21 Spaces: 4 UTF-8 CRLF Python 3.13.6 Go Live ENG FR 3:51 PM 2/15/2026

Administrator: Windows PowerShell

PS C:\Windows\system32> netstat -nb

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:9500	127.0.0.1:64210	CLOSE_WAIT
[python.exe]			
TCP	127.0.0.1:63267	127.0.0.1:63268	ESTABLISHED
[nbcode64.exe]			
TCP	127.0.0.1:63268	127.0.0.1:63267	ESTABLISHED
[Code.exe]			
TCP	127.0.0.1:64210	127.0.0.1:9500	FIN_WAIT_2
[System]			
TCP	172.23.15.215:51214	23.53.1.146:443	CLOSE_WAIT
[SearchApp.exe]			
TCP	172.23.15.215:60319	41.111.227.35:53	TIME_WAIT
TCP	172.23.15.215:60321	23.53.1.134:443	LAST_ACK
[msedge.exe]			
TCP	172.23.15.215:60322	2.22.5.114:80	ESTABLISHED
CryptSvc			
[svchost.exe]			
TCP	172.23.15.215:60323	13.107.213.43:443	CLOSE_WAIT
[Code.exe]			
TCP	172.23.15.215:60324	13.107.213.43:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:60325	142.250.178.3:80	ESTABLISHED
CryptSvc			
[svchost.exe]			
TCP	172.23.15.215:60327	23.53.1.134:443	ESTABLISHED
[msedge.exe]			
TCP	172.23.15.215:60331	20.190.177.149:443	SYN_SENT
wlidsvc			
[svchost.exe]			
TCP	172.23.15.215:61877	98.66.133.184:443	ESTABLISHED
WpnService			
[svchost.exe]			
TCP	172.23.15.215:63259	74.242.255.116:443	ESTABLISHED
[msedge.exe]			
TCP	172.23.15.215:64184	52.168.117.168:443	FIN_WAIT_2

Administrator: Windows PowerShell

PS C:\Windows\system32> netstat -n -b

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:9500	127.0.0.1:51233	ESTABLISHED
[python.exe]			
TCP	127.0.0.1:9500	127.0.0.1:57428	ESTABLISHED
[python.exe]			
TCP	127.0.0.1:51233	127.0.0.1:9500	ESTABLISHED
[python.exe]			
TCP	127.0.0.1:57428	127.0.0.1:9500	ESTABLISHED
[java.exe]			
TCP	172.23.15.215:51075	74.242.255.116:443	ESTABLISHED
[msedge.exe]			
TCP	172.23.15.215:51214	23.53.1.146:443	CLOSE_WAIT
[SearchApp.exe]			
TCP	172.23.15.215:51231	52.168.112.66:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:51241	23.38.167.18:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57388	23.53.1.145:443	LAST_ACK
[msedge.exe]			
TCP	172.23.15.215:57389	150.171.74.16:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57392	23.203.61.40:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57393	23.203.61.40:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57394	23.203.61.40:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57395	23.203.61.40:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57396	23.203.61.40:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57397	23.203.61.10:443	ESTABLISHED
[Code.exe]			
TCP	172.23.15.215:57398	23.203.61.10:443	ESTABLISHED
[Code.exe]			



ENG 3:37 PM
FR 2/15/2026



Djalil



```
Administrator: Windows PowerShell

PS C:\Windows\system32> netstat -nao |findstr :9500
TCP    127.0.0.1:9500          0.0.0.0:0              LISTENING          11956
TCP    127.0.0.1:9500          127.0.0.1:51184         ESTABLISHED        11956
TCP    127.0.0.1:51184        127.0.0.1:9500         ESTABLISHED        9400

PS C:\Windows\system32> taskkill /PID 11956 /F ; taskkill /PID 9400 /F
SUCCESS: The process with PID 11956 has been terminated.
SUCCESS: The process with PID 9400 has been terminated.
PS C:\Windows\system32>
```

Capturing from Adapter for loopback traffic capture

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 9500

No.	Time	Source	Destination	Protocol	Length	Info
14	30.649236900	127.0.0.1	127.0.0.1	TCP	56	51184 → 9500 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
15	30.649315300	127.0.0.1	127.0.0.1	TCP	56	9500 → 51184 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
16	30.649375400	127.0.0.1	127.0.0.1	TCP	44	51184 → 9500 [ACK] Seq=1 Ack=1 Win=2619648 Len=0
17	34.359691000	127.0.0.1	127.0.0.1	TCP	51	51184 → 9500 [PSH, ACK] Seq=1 Ack=1 Win=2619648 Len=7
18	34.359760200	127.0.0.1	127.0.0.1	TCP	44	9500 → 51184 [ACK] Seq=1 Ack=8 Win=2619648 Len=0

Frame 14: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NPF{...} Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 51184, Dst Port: 9500, Seq: 0, Len: 0

0000 02 00 00 00 45 00 00 34 3a 04 40 00 80 06 00 00E 4 :@.....
0010 7f 00 00 01 7f 00 00 01 c7 f0 25 1c 78 01 91 15%x.....
0020 00 00 00 00 80 02 ff ff 80 c6 00 00 02 04 ff d7
0030 01 03 03 08 01 01 04 02

Packets: 18 · Displayed: 5 (27.8%) Profile: Default

ENG 3:15 PM
FR 2/15/2026

```
File Edit Selection View Go Run ...  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\Djalil\Documents\Gith  
ub\cs-mat06\Reseaux\TP> python -u  
erials\Semestre06\Reseaux\TP> pyt  
hon -u "crveur.py"  
PS C:\Users\Djalil\Documents\GitHub\cs-materials\Semestre06\Resea  
y"  
Lancement serveur  
Connexion de la machine = ('127.0.0.1', 59669)  
Je suis la  
Message  
█
```

```
Python  
PS C:\Users\Djalil\Documents\Gith  
ub\cs-mat06\Reseaux\TP> & C:/User  
erials\Semestre06\Reseaux\TP> & C  
:/Users/Dlil/Documents/GitHub/cs-  
jalil/AppData/Local/Programs/Pyth  
Console Client  
Je suis la  
Message  
█  
main* 0.1.61 0 0 0  
ENG 2:45 PM  
FR 2/15/2026
```

FichierSortie.txt - Notepad

File Edit Format View Help

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:9500	127.0.0.1:59669	ESTABLISHED
TCP	127.0.0.1:59669	127.0.0.1:9500	ESTABLISHED
TCP	172.23.15.215:52885	98.66.133.184:443	ESTABLISHED
TCP	172.23.15.215:52926	23.53.1.146:443	CLOSE_WAIT
TCP	172.23.15.215:52927	23.53.1.146:443	CLOSE_WAIT
TCP	172.23.15.215:52928	23.53.1.146:443	CLOSE_WAIT
TCP	172.23.15.215:52932	52.98.243.146:443	ESTABLISHED
TCP	172.23.15.215:52933	52.98.243.146:443	ESTABLISHED
TCP	172.23.15.215:52936	23.53.1.156:443	CLOSE_WAIT
TCP	172.23.15.215:52937	23.53.1.156:443	ESTABLISHED
TCP	172.23.15.215:52938	23.53.1.156:443	ESTABLISHED
TCP	172.23.15.215:52939	23.53.1.156:443	CLOSE_WAIT
TCP	172.23.15.215:52940	23.53.1.146:443	CLOSE_WAIT
TCP	172.23.15.215:52996	74.242.255.116:443	ESTABLISHED
TCP	172.23.15.215:57846	13.107.213.43:443	CLOSE_WAIT
TCP	172.23.15.215:57864	150.171.27.11:443	ESTABLISHED
TCP	172.23.15.215:57865	150.171.27.11:443	ESTABLISHED
TCP	172.23.15.215:57866	20.42.73.28:443	ESTABLISHED

Ln 1, Col 1 100% Windows (CRLF) UTF-16 LE

